

Project Documentation

Project: EduGenie: Google Gemini Powered Learning Assistant

Team ID	Team Size	Team Leader	Team Members
SWTID-2026-5228	5	Madhubala G	Pavithra G, Logeswari G, Sivapriya W, Gopika M

1. Introduction

EduGenie is a learning assistant that uses Google Gemini to help students understand topics, revise quickly and practice for exams. A student can ask a doubt in simple language, paste notes to get a short summary, generate a quiz on any topic, or receive a day-wise study plan. The application is built with Python and Streamlit and runs in a web browser.

2. Problem Statement

Many students do not get instant, personalized help when they are stuck on a concept. Textbooks are dense, online resources are scattered, and preparing notes, quizzes and plans by hand takes a lot of time. This leaves students confused, anxious and inconsistent in their preparation.

3. Objectives

- Give instant, simple explanations for student doubts at their chosen level.
- Convert long notes into short, revision-friendly summaries.
- Generate practice quizzes with answers and explanations.
- Create a personalized day-wise study plan.
- Provide a clean, easy-to-use interface that any student can use.

4. Team Details

Role	Name
Team ID	SWTID-2026-5228
Team Size	5
Team Leader	Madhubala G
Team Member	Pavithra G
Team Member	Logeswari G
Team Member	Sivapriya W
Team Member	Gopika M

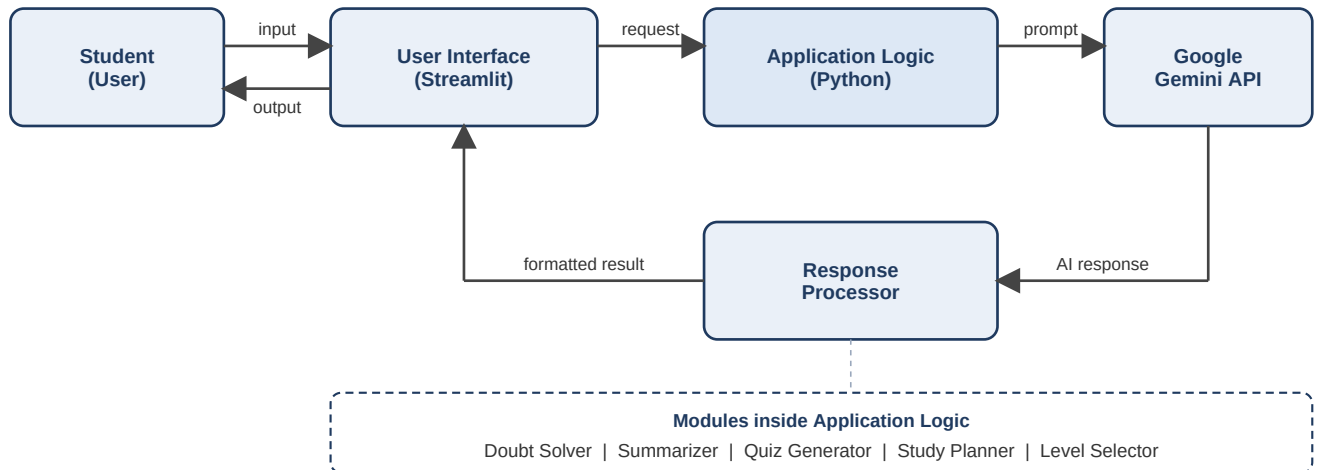
5. Proposed Solution and Features

EduGenie combines four learning tools in one application. Each tool sends a well-designed prompt to the Gemini model and shows the result to the student.

Feature	Description
Doubt Solver	Explains a question at beginner, intermediate or advanced level
Notes Summarizer	Turns long notes into short bullet points
Quiz Generator	Creates 3 to 10 MCQs with correct answers and explanations

Feature	Description
Study Planner	Builds a day-wise plan from subject, days and hours per day
Chat History	Keeps the questions of the current session in the sidebar

6. System Architecture



The student enters input in the Streamlit interface. The application logic validates the input, builds a prompt and calls the Gemini API. The response is formatted and displayed back to the student.

7. Technology Used

Layer	Technology
Frontend	Streamlit
Backend	Python
AI Model	Google Gemini API (google-genai SDK)
Configuration	python-dotenv (.env file)
Version Control	Git and GitHub

8. Implementation

The project has two main files. `gemini_utils.py` contains one reusable function that talks to Gemini and four small functions that build the prompts for each feature. `app.py` contains the Streamlit interface with four tabs, input validation, a loading spinner and the session chat history. The API key is read from a `.env` file so it is never written in the code.

[Insert screenshot: Home screen with the four tabs]

[Insert screenshot: Doubt Solver answer]

[Insert screenshot: Generated quiz]

9. Testing and Results

The application was tested with functional test cases for each module and with error cases such as empty input and an invalid API key. Detailed test cases and results are given in the Project Testing documents. The summary is filled in after testing.

Parameter	Value
Total test cases	11
Passed	[fill]
Failed	[fill]
Average response time	[fill]

10. Conclusion

EduGenie shows how a generative AI model can act as a personal learning assistant for students. By bringing doubt solving, summarizing, quizzing and planning into one simple application, it saves study time and supports self-paced learning.

11. Future Scope

- Voice input and spoken answers.
- PDF and image upload for notes.
- Answers in more Indian languages, including Tamil.
- Progress tracking and a score dashboard.
- Login and saved history across sessions.
- Dashboards for teachers and institutions.